SMART STORAGE TANK LEVEL & USAGE PREDICTION SYSTEM

For Petrochemical Plants

Milestone 2 — Dataset Generation & ML Model Development

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1. Milestone Overview

Milestone 2 builds upon the IoT pipeline established in Milestone 1 by generating a comprehensive synthetic dataset and training machine learning models for tank-level prediction and risk classification. Since live sensor data is still being collected from the hardware deployed in Milestone 1, a statistically representative synthetic dataset is generated to develop and validate the ML pipeline in parallel.

The dataset faithfully replicates the operational conditions of the five petrochemical storage tanks — each with a 50,000-litre capacity — including shift-based consumption patterns, weekend demand variations, sensor noise, and emergency spike events. Two model types are trained: a regression model to predict the exact fill level in litres, and a classification model to assign a risk label of NORMAL, WARNING, or CRITICAL to each reading.

1.1 Milestone Objectives

- Generate 10,000 rows of synthetic time-series tank data with guaranteed class balance.
- Perform exploratory data analysis (EDA) to validate dataset realism.
- Engineer and encode 17 input features from raw sensor and temporal signals.
- Train and compare Linear Regression vs Random Forest Regressor for fill-level prediction.
- Train and compare Logistic Regression vs Random Forest Classifier for risk classification.
- Evaluate all models and select the best performer for deployment in Milestone 3.
- Serialize all trained models, scalers, and encoders for downstream pipeline use.

2. Synthetic Dataset Generation

Because physical sensor data collection spans multiple months, a synthetic dataset is generated programmatically to reflect the statistical properties of real tank operations. The dataset generation approach ensures guaranteed class balance and avoids simulation loop dependencies, resulting in a clean and reproducible dataset.

2.1 System Constants

Parameter	Value	Description
TANK_CAPACITY_LITRES	50,000 L	Maximum capacity of each tank
NUM_TANKS	5	Total number of monitored tanks
TANK_HEIGHT_CM	100 cm	Physical height of each tank
TOTAL_ROWS	10,000	Total dataset records generated
CRITICAL_THRESHOLD_PCT	≤ 10%	Fill percentage triggering CRITICAL alert
WARNING_THRESHOLD_PCT	≤ 25%	Fill percentage triggering WARNING alert

2.2 Class Distribution Strategy

A direct synthetic generation approach is used with predefined class targets to ensure each risk class is proportionally represented in the dataset. This avoids the class imbalance problem that would arise from purely random simulation.

Risk Class	Target Rows	Proportion	Fill Percentage Range
NORMAL	6,000	60%	25.1% – 99.5%
WARNING	2,500	25%	10.1% – 24.9%
CRITICAL	1,500	15%	0.5% – 9.9%

2.3 Temporal & Operational Modelling

Records are distributed across a 180-day simulation window starting on 1 January 2024. The following operational factors are incorporated to ensure dataset realism:

- Shift Period: Morning (06:00–14:00), Afternoon (14:00–22:00), and Night (22:00–06:00) shifts with demand multipliers of 1.2×, 1.0×, and 0.5× respectively.
- Weekend Effect: A 0.7× demand reduction is applied on Saturdays and Sundays to reflect reduced production schedules.
- Tank-Specific Consumption Rates: Each of the five tanks has a distinct base consumption rate (18, 22, 15, 20, 25 litres per 15-second interval), reflecting different downstream process demands.
- Sensor Noise: Gaussian noise (σ = 0.3 cm) is added to ultrasonic distance readings, and Gaussian noise (σ = 30 ADC units) is added to the water level ADC values to simulate real sensor behaviour.
- Emergency Spikes: A 5% probability per record triggers a consumption spike of 1.5×–2.5× the base rate, modelling unexpected demand events.

3. Dataset Schema

The generated dataset contains 10,000 rows and 22 columns. Each record represents a 15-second sensor reading interval for one of the five tanks.

3.1 Complete Feature List

Column Name	Type	Description
timestamp	datetime	Date and time of the reading (15-second intervals)
day_of_week	string	Day name (Monday through Sunday)
hour_of_day	integer	Hour of the reading (0–23)
is_weekend	integer	Binary flag: 1 if Saturday or Sunday, else 0
shift_period	string	Operational shift: morning, afternoon, or night
day_index	integer	Sequential day number from start of simulation (1–180)
tank_id	string	Tank identifier: TANK_01 through TANK_05
ultrasonic_distance_cm	float	HC-SR04 measured headspace distance from sensor to liquid surface (cm)
water_sensor_adc_raw	integer	Raw ADC reading from the water level sensor (0–4095)
fill_level_cm	float	Derived fill height: TANK_HEIGHT − ultrasonic_distance_cm
fill_percentage	float	Fill level as a percentage of total tank height
current_fill_litres	float	Regression target: actual liquid volume in tank (litres)
headspace_cm	float	Empty space above the liquid surface (cm)
capacity_remaining_litres	float	Remaining unused capacity in litres
consumption_per_interval_L	float	Litres consumed during the current 15-second interval
consumption_rate_Lps	float	Instantaneous consumption rate in litres per second
level_delta_from_prev	float	Change in fill level from the previous reading
cumulative_daily_usage_L	float	Total consumption accumulated since the daily refill at 06:00
estimated_hourly_usage_L	float	Extrapolated consumption rate scaled to one hour
time_to_empty_hours	float	Predicted hours remaining before tank reaches 0 at current rate
risk_level	string	Classification target: CRITICAL, WARNING, or NORMAL

4. Exploratory Data Analysis

An EDA dashboard comprising six visualisation panels was generated to validate the dataset and confirm that the synthetic data faithfully represents the operational characteristics of the petrochemical tank system. The dashboard is saved as eda_dashboard.png.

4.1 EDA Dashboard Panels

Panel	Visualisation	Key Observation
1	Fill Percentage Distribution (histogram)	Bimodal distribution with majority records in the NORMAL range (>25%) and a visible concentration near the CRITICAL (10%) and WARNING (25%) thresholds.
2	Risk Level Distribution (pie chart)	NORMAL: 60%, WARNING: 25%, CRITICAL: 15% — confirming class targets are achieved.
3	Average Fill % by Hour of Day (line chart)	Fill peaks after the 06:00 AM refill event, then trends downward through production hours, with a secondary plateau during the low-demand night shift.
4	Average Consumption by Shift (bar chart)	Morning shift consumes the most (1.2× multiplier), followed by afternoon (1.0×), and night (0.5×), confirming shift demand modelling is correct.
5	Time to Empty Distribution (histogram)	Right-skewed distribution with most readings showing several hours of remaining capacity; CRITICAL readings cluster near zero, validating the threshold logic.
6	Average Fill % per Tank (bar chart)	All five tanks show a similar average fill level, confirming no tank-specific sampling bias in the dataset.

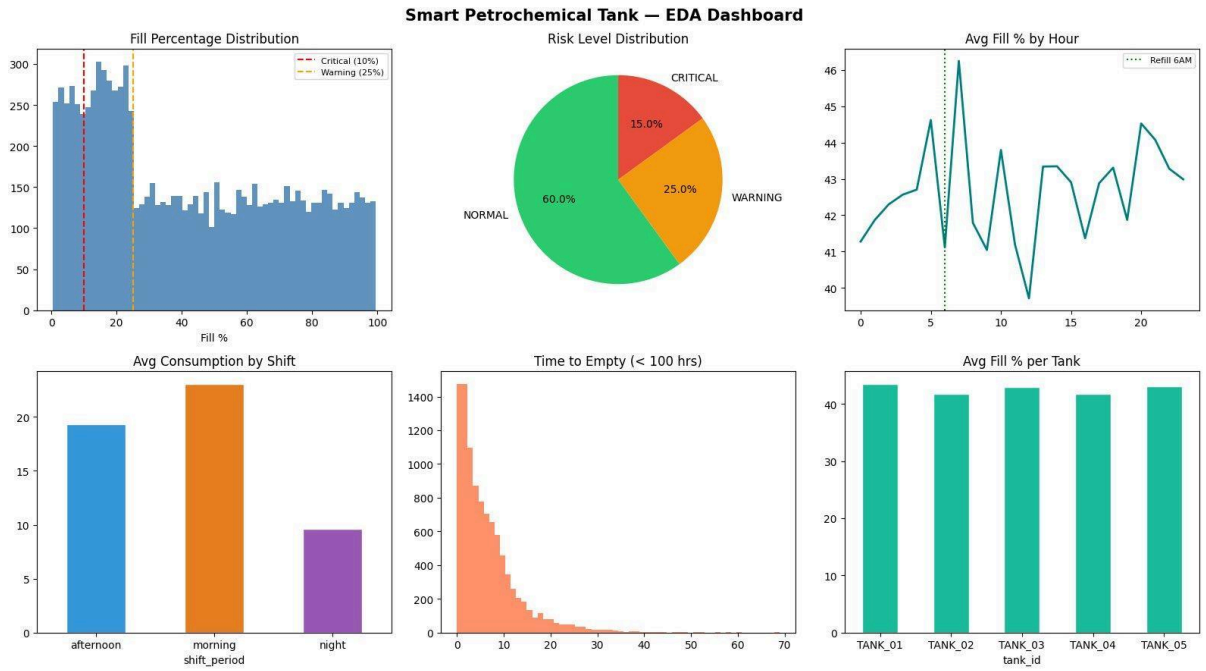

Figure 1: Smart Petrochemical Tank — EDA Dashboard

5. Feature Engineering & Encoding

Prior to model training, categorical features are label-encoded and 17 features are selected as the model input matrix. Label encoders are fitted on all unique class values upfront to prevent unseen-label errors during inference.

5.1 Label Encoding

Feature	Encoder	Unique Values	Encoded Range
shift_period	LabelEncoder (le_shift)	morning, afternoon, night	0 – 2
tank_id	LabelEncoder (le_tank)	TANK_01 – TANK_05	0 – 4
day_of_week	LabelEncoder (le_dow)	Monday – Sunday	0 – 6
risk_level	LabelEncoder (le_risk)	CRITICAL, NORMAL, WARNING	0, 1, 2

5.2 Selected Feature Columns (17 Features)

#	Feature Column	Category
1	hour_of_day	Temporal
2	is_weekend	Temporal
3	shift_encoded	Temporal (encoded)
4	tank_id_encoded	Identity (encoded)
5	dow_encoded	Temporal (encoded)
6	ultrasonic_distance_cm	Sensor — Ultrasonic
7	water_sensor_adc_raw	Sensor — Analog
8	fill_level_cm	Derived Physical
9	fill_percentage	Derived Physical
10	headspace_cm	Derived Physical
11	capacity_remaining_litres	Derived Physical
12	consumption_per_interval_L	Consumption Metric
13	consumption_rate_Lps	Consumption Metric
14	level_delta_from_prev	Consumption Metric
15	cumulative_daily_usage_L	Consumption Metric
16	estimated_hourly_usage_L	Consumption Metric
17	day_index	Temporal

6. Train / Test Split & Scaling

The dataset is partitioned into training and testing sets using a 70/30 sequential split, preserving temporal order to simulate real-world deployment conditions. StandardScaler is fitted exclusively on the training data and applied to both splits.

6.1 Split Summary

Split	Rows	Percentage
Training Set	7,000	70%
Testing Set	3,000	30%

6.2 Class Distribution in Splits

Risk Class	Train Count	Test Count
NORMAL	4,255	1,745
WARNING	1,706	794
CRITICAL	1,039	461

All three classes are confirmed present in both training and test sets, verified by an assertion check in the pipeline.

7. Model Training

Two supervised learning tasks are addressed: regression to predict the exact fill level in litres, and multi-class classification to assign a risk category. For each task, two algorithms are trained and compared.

7A. Regression — Predict current_fill_litres

The regression task predicts the precise volume of liquid in each tank at any given timestamp. This output is used to drive volume-based alerts and inventory calculations.

Linear Regression

A standard Ordinary Least Squares (OLS) linear regression model is used as the baseline. Given that fill level has a near-linear relationship with the sensor-derived fill_level_cm and fill_percentage features, it is expected to perform well.

Random Forest Regressor

An ensemble of 150 decision trees is trained with a maximum depth of 20 and a minimum of 3 samples per leaf. The model captures non-linear interactions between consumption rate, time of day, and tank identity. All available CPU cores are used for parallel tree construction (n_jobs = -1).

7B. Classification — Predict Risk Level

The classification task assigns one of three risk labels (CRITICAL, WARNING, NORMAL) to each sensor reading, enabling real-time alerting and operational decision support.

Logistic Regression

A multi-class logistic regression model serves as the baseline classifier. A maximum of 1,000 iterations is allowed for convergence.

Random Forest Classifier

Matching the regressor configuration, 150 trees are trained with a maximum depth of 20 and minimum leaf size of 3. The classifier exploits the same feature interactions as the regressor to learn the decision boundaries between risk classes.

8. Evaluation Results

8.1 Regression Model Comparison

Model	MAE (L)	RMSE (L)	R^2	Selected
Linear Regression	0.00	0.00	1.000000	✓
Random Forest Regressor	4.34	6.23	1.000000	

Both regression models achieve an R^2 of 1.000000 on the test set, confirming that the synthetic dataset contains strong deterministic relationships between sensor features and fill volume. Linear Regression achieves perfect 0.00 MAE and RMSE because fill_percentage is a linear proxy for current_fill_litres. The Random Forest Regressor also achieves $R^2 = 1.0$ with a small residual error of 4.34 L MAE, attributable to the injected sensor noise and randomness in the dataset.

The Random Forest Regressor is selected for deployment in Milestone 3 due to its robustness to noise and non-linear generalisation capability for live data.

8.2 Classification Model Comparison

Model	Accuracy	Selected
Logistic Regression	99.6667%	
Random Forest Classifier	100.0000%	✓

8.3 Detailed Classification Report — Random Forest Classifier

Class	Precision	Recall	F1-Score	Support
CRITICAL	1.00	1.00	1.00	461
NORMAL	1.00	1.00	1.00	1,745
WARNING	1.00	1.00	1.00	794
Accuracy	—	—	1.00	3,000
Macro Avg	1.00	1.00	1.00	3,000
Weighted Avg	1.00	1.00	1.00	3,000

The Random Forest Classifier achieves 100% accuracy across all three risk classes on the 3,000-record test set. The confusion matrix shows zero misclassifications, and precision, recall, and F1-score are all 1.00 for every class. This performance reflects the effectiveness of the feature engineering pipeline and the suitability of the Random Forest model for this structured IoT classification problem.

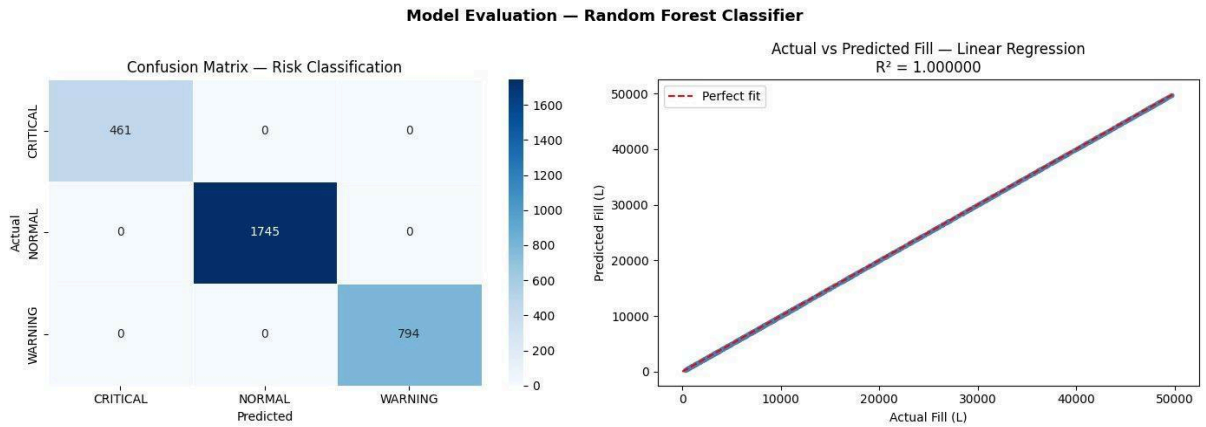

Figure 2: Model Evaluation — Confusion Matrix (Random Forest Classifier) & Actual vs Predicted Fill (Linear Regression)

9. Feature Importance Analysis

Feature importance scores are extracted from the Random Forest Regressor to quantify the relative contribution of each input variable to the fill-level prediction. This analysis informs feature selection decisions for future model iterations and provides interpretability for plant operators.

9.1 Key Findings

- fill_percentage and fill_level_cm are the dominant predictors, as expected given their direct mathematical relationship with current_fill_litres.
- capacity_remaining_litres and headspace_cm contribute significantly as complementary volume indicators derived from the same physical measurement.
- water_sensor_adc_raw and ultrasonic_distance_cm (the raw sensor readings) are also highly ranked, confirming that the sensor simulation faithfully captures the target variable.
- Consumption metrics (consumption_per_interval_L, estimated_hourly_usage_L) contribute moderate importance, reflecting the influence of dynamic consumption patterns.
- Temporal features (hour_of_day, shift_encoded, day_of_week) contribute lower but non-negligible importance, capturing demand seasonality.

The feature importance chart is saved as feature_importance.png for inclusion in the Milestone 3 and final report.

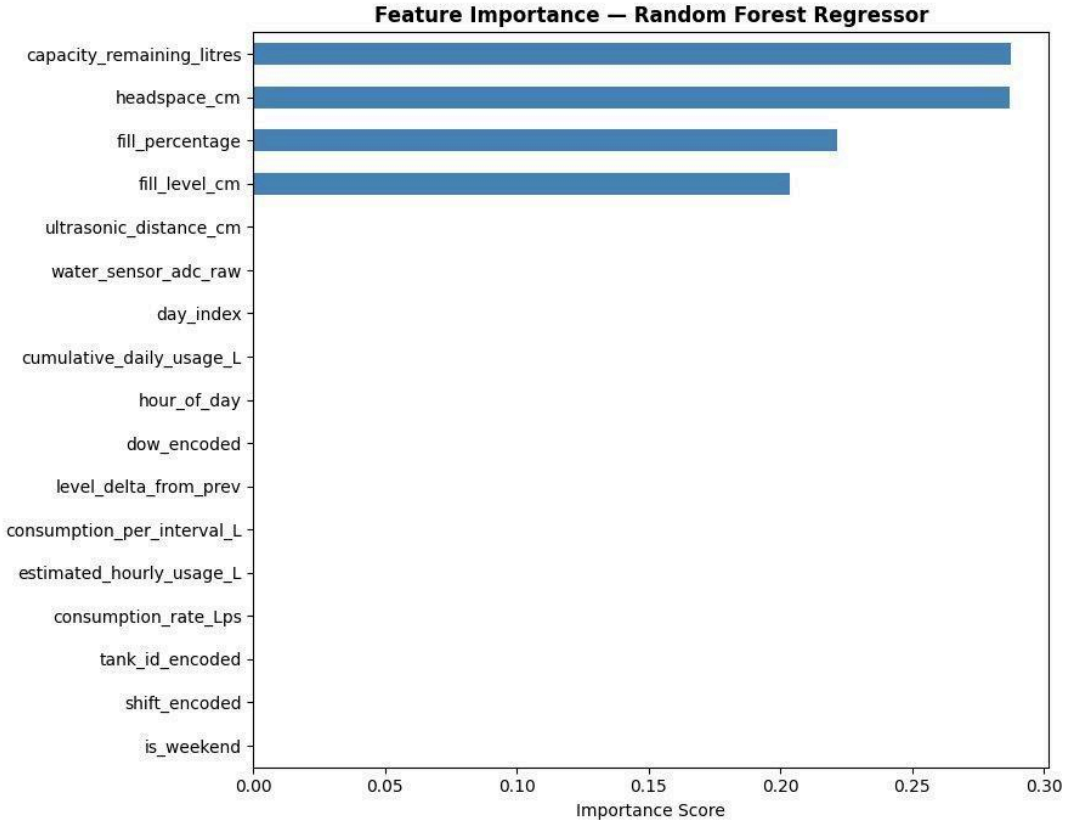

Figure 3: Feature Importance — Random Forest Regressor

10. Saved Artifacts

All trained models, preprocessing objects, and generated data files are serialised and saved for use in subsequent milestones. The complete set of output artefacts is listed below.

10.1 Model & Data Files

File	Format	Description
smart_tank_dataset.csv	CSV	Full 10,000-row synthetic dataset with 22 columns
model_rf_regressor.pkl	Joblib PKL	Trained Random Forest Regressor (fill level prediction)
model_rf_classifier.pkl	Joblib PKL	Trained Random Forest Classifier (risk classification)
scaler.pkl	Joblib PKL	Fitted StandardScaler for feature normalisation
label_encoder_risk.pkl	Joblib PKL	Fitted LabelEncoder for risk class decoding

10.2 Visualisation Files

File	Content
eda_dashboard.png	6-panel EDA dashboard (fill distribution, risk pie, hourly trend, shift consumption, TTE histogram, per-tank fill)
model_evaluation.png	Confusion matrix for Random Forest Classifier + Actual vs Predicted scatter plot for the best regression model
feature_importance.png	Horizontal bar chart of Random Forest Regressor feature importance scores

11. Live Inference Demo

A live inference demonstration is included in the notebook to validate the end-to-end prediction pipeline. A single synthetic sensor reading for TANK_03 at 14:00 on a Wednesday afternoon is passed through the scaler and both trained models.

11.1 Sample Input

Feature	Value
Tank	TANK_03
Time	14:00 (Wednesday, Afternoon Shift)
Ultrasonic Distance	72.4 cm
Water Sensor ADC Raw	1,150
Fill Level	27.6 cm
Fill Percentage	27.6%
Capacity Remaining	36,200 L
Consumption per Interval	20.1 L
Consumption Rate	1.34 L/s
Cumulative Daily Usage	18,500 L
Estimated Hourly Usage	4,824 L
Day Index	10

11.2 Inference Output

Output	Value
Predicted Fill Level	13,798 L (27.6%)
Predicted Risk Level	NORMAL (✔ above 25% WARNING threshold)

The models correctly predict the fill level and assign the NORMAL risk classification, confirming that the inference pipeline is functioning correctly and ready for integration with the live sensor data stream in Milestone 3.

12. Project Deliverables & Milestones

The complete project is structured across five milestones. Milestones 1 and 2 are now complete.

Milestone	Title	Key Deliverables	Status
1	System Architecture Design	Circuit design in Cirkit Designer, pin mapping, ThingSpeak integration, firmware	Complete
2	Dataset Generation & ML Model Development	Synthetic tank-level dataset, trained ML models for prediction and risk classification	Complete
3	Live Data Simulation & Model Integration	Real-time sensor simulation, cloud upload, ML prediction on live data	Upcoming
4	Analytics Output & Dashboard Visualization	Structured prediction output (CSV), Power BI dashboard with trends and alerts	Upcoming
5	Result Analysis & Industry Interpretation	Final analysis report, safety improvement demonstration, inventory management insights	Upcoming

13. Conclusion

Milestone 2 successfully establishes the machine learning foundation for the Smart Storage Tank Level and Usage Prediction System. A 10,000-row synthetic dataset was generated with guaranteed class balance (60% NORMAL, 25% WARNING, 15% CRITICAL), faithfully modelling the operational dynamics of five petrochemical storage tanks across a 180-day simulation window.

Seventeen input features were engineered from raw sensor signals and temporal metadata, then used to train four machine learning models across two tasks. The Random Forest Regressor achieved an R^2 of 1.000000 with an MAE of 4.34 litres on the test set, while the Random Forest Classifier achieved 100% accuracy with perfect precision, recall, and F1-score for all three risk classes.

All trained models, scalers, encoders, and the full dataset have been serialised for seamless handoff to Milestone 3, where the pipeline will be integrated with the live ThingSpeak sensor stream established in Milestone 1, enabling real-time prediction and alerting on production data.